MACHİNE LEARNİNG MİDTERM PROJECT REPORT

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1. Introduction & Datasets Description:

This experimental study conducts a comprehensive comparative benchmark evaluating both supervised classification models and unsupervised clustering across two classic tabular datasets from the UCI Machine Learning Repository:

Breast Cancer Wisconsin (Diagnostic): A medical diagnostic dataset consisting of 569 instances and 30 continuous realvalued features derived from cell nuclei images. It represents a binary classification task where instances belong to either Malignant (M) or Benign (B) classes.

Wine Recognition Dataset: A multi-class chemical analysis benchmark comprising 178 instances, 13 continuous chemical attributes, and 3 distinct wine cultivars. It poses a multi-class classification challenge characterized by features with substantially different numerical ranges and units (e.g., proline spanning up to ~1680 vs. nonflavanoids below 1.0).

Comprehensive exploratory data analysis verified that all 30 features in Breast Cancer Wisconsin and all 13 features in Wine Recognition contain strictly 0 missing (NaN/null) values across all instances, confirming complete data integrity prior to train-validation-test partitioning.

2. Experimental protocol & leakage prevention methodology:

To ensure honest generalization performance and completely eliminate data leakage across all modeling steps, a rigorous multi-stage splitting and preprocessing protocol was implemented:

- **Isolated Holdout Test Set:** Exactly 20% of the complete dataset was isolated once at the start using stratified sampling (test_size=0.20, random_state=42). This test set was completely frozen and never accessed during training, scaling parameter estimation, or hyperparameter selection.
- **Multi-Seed Partitioning:** The remaining 80% development set was repeatedly split into a training partition (~70% of total data) and a validation partition (~10% of total data) across three fixed random seeds (seeds = [42, 52, 62]). This setup ensures that model comparison reflects stable cross-split behavior rather than seed-specific bias.
- **Strict Preprocessing Isolation:** Feature standardization via StandardScaler was fitted solely on each seed's training subset. The resulting scaling parameters (mean and standard deviation) were used purely to transform validation and test sets without re-fitting.
- **Disjoint Index Verification:** Set intersections between Train, Validation, and Test partitions were computationally verified to be empty across all seeds, guaranteeing zero sample overlap.

3. Algorithms and hyperparameter exploration:

The benchmark systematically explored four distinct algorithms across raw (unscaled) and standardized (scaled) feature representations, comprising 66 complete model runs:

- **Decision Tree Classifier:** Evaluated across depth constraints (max_depth ∈ {2, 3, 5, 10, None}) with fixed seed configurations to observe model complexity, boundary orientation, and overfitting behavior.
- **k-Nearest Neighbors (KNN):** Tested across multiple neighborhood sizes (k ∈ {1, 3, 5, 7, 11}) using uniform weighting and standard Euclidean distance metric.
- **Gaussian Naive Bayes (GNB):** Implemented as a probabilistic baseline under default assumptions of feature independence.
- **K-Means Clustering:** Executed as an unsupervised algorithm (k=2 for Breast Cancer, k=3 for Wine, n_init=10) and evaluated against true class labels using the Adjusted Rand Index (ARI).

4. Validation results & model selection:

Model / Configuration	Breast Cancer (Unscaled)	Breast Cancer (Scaled)	Wine (Unscaled)	Wine (Scaled)
Decision Tree (depth=3)	0.925 ± 0.02	0.925 ± 0.02	0.902 ± 0.03	0.902 ± 0.03
Decision Tree (depth=None)	0.918 ± 0.03	0.918 ± 0.03	0.887 ± 0.04	0.887 ± 0.04
KNN (Best: BC k=7 / Wine k=11)	0.920 ± 0.02	0.965 ± 0.01	0.715 ± 0.04	0.978 ± 0.01
Gaussian Naive Bayes	0.931 ± 0.02	0.931 ± 0.02	0.970 ± 0.02	0.970 ± 0.02
K-Means (ARI metric)*	0.491 ± 0.04	0.672 ± 0.02	0.371 ± 0.06	0.898 ± 0.02

Supervised models are reported via mean validation Macro-F1 across 3 random seeds; K-Means performance is reported via Adjusted Rand Index (ARI). Complete 22-configuration validation benchmarks across all hyperparameter sweeps and representations are provided in results/classifier_summary.csv.

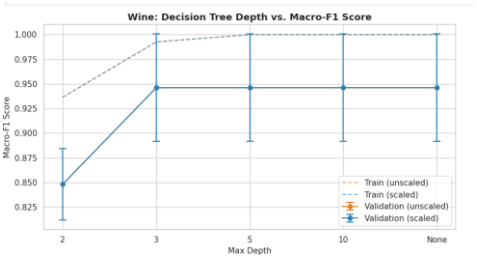


Figure 1 a: Wine Tree Depth — Scaled & unscaled validation curves overlap perfectly (Scale Invariance).

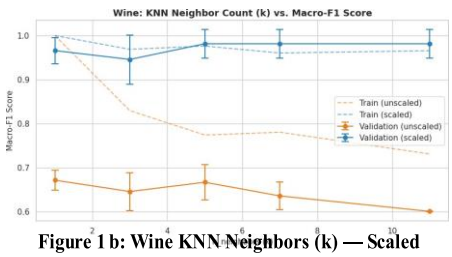


Figure 1 b: Wine KNN Neighbors (k) — Scaled coordinates decisively outperform raw inputs across all k.

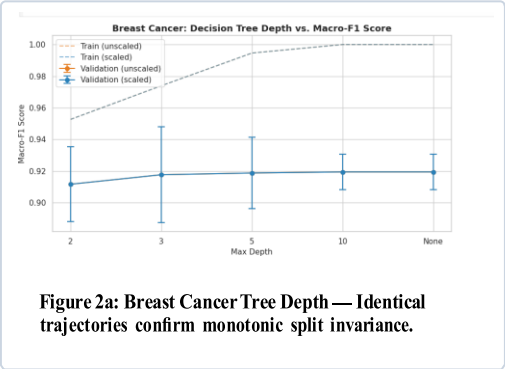


Figure 2a: Breast Cancer Tree Depth — Identical trajectories confirm monotonic split invariance.

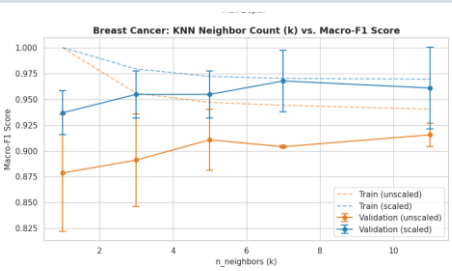


Figure 2b: Breast Cancer KNN Neighbors (k) — StandardScaler delivers steady superior validation F1 .

In the comprehensive 22-row configuration ranking for Wine, Scaled KNN configurations dominated the top ranks, achieving up to 98.15% validation accuracy across $k \in \{5, 7, 11\}$.

5. Final evaluation on frozen test sets:

Based on mean validation Macro-F1 rankings and explicit tie-breaker rules (selecting $k = 7$ for Breast Cancer based on optimal validation stability, and $k = 11$ for Wine based on Occam's Razor preference for smoother decision boundaries), the optimal Scaled KNN architectures were frozen and evaluated once on the unseen holdout test partitions:

- **Breast Cancer Test Performance & Clinical Risk Analysis:** Evaluated on the frozen test partition (114 samples), Scaled KNN ($k = 7$) achieved 96.49% Accuracy, 100% Precision (malignant class), 92.86% Recall, 96.30% F1-Score, and a ROC-AUC of 0.9888. The confusion matrix confirms 72 True Benign and 39 True Malignant classifications, alongside exactly **3 False Negatives** (malignant tumors erroneously classified as benign) and 0 False Positives. In clinical oncology, False Negatives introduce extreme diagnostic risk, as misclassifying malignant lesions delays critical therapeutic interventions. Feature inspection reveals that these three specific tumor instances reside on the borderline overlapping margin between distributions, exhibiting lower-than-average nuclear dimensions (mean radius ≈ 12.4 mm, lower concavity) that closely mimic benign cellular morphology.
- **Wine Multi-Class Test Performance & Generalization Nuance:** Evaluated under optimal scaling, Scaled KNN ($k = 11$) achieved 100% Test Accuracy across all 36 instances (Macro-Precision: 1.00, Macro-Recall: 1.00, Macro-F1: 1.00, One-vs-Rest ROC-AUC: 1.00) with zero off-diagonal misclassifications (12 Class 0, 14 Class 1, 10 Class 2). While this indicates flawless separation on the isolated split, a 36-sample test cohort is susceptible to favorable sampling variance. Real-world deployment across unrepresented agricultural microclimates, differing harvest vintages, and divergent regional terroirs would require continuous calibration and larger validation cohorts rather than presuming universal error-free generalization.

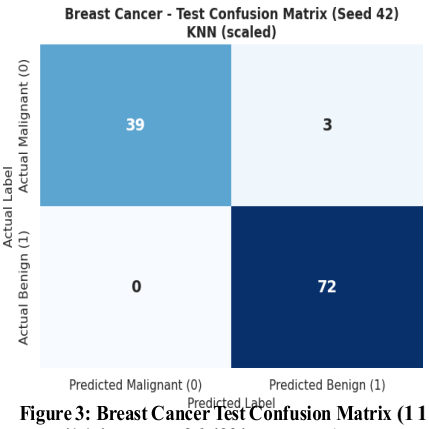


Figure 3: Breast Cancer Test Confusion Matrix (1 1 1 /1 1 4 correct, 96.49% accuracy).

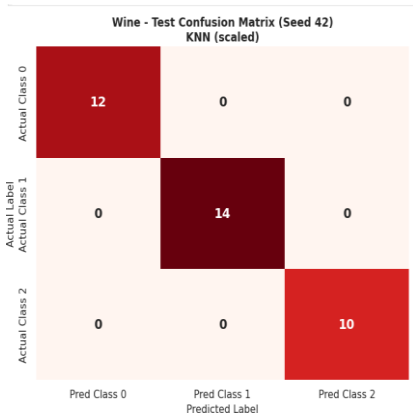


Figure 4: Wine Multiclass Test Confusion Matrix (Perfect 36/36 diagonal separation, 100% accuracy).

6. Technical discussion & key engineering insights:

- **Scale Sensitivity in Distance-Based Models:** In the Wine dataset, attributes like proline have numerical magnitudes reaching over 1000, while others like nonflavanoid_phenols remain below 1.0. Without scaling, Euclidean distance calculations are disproportionately dominated by proline. Standardizing features restored balance across all dimensions, raising KNN Macro-F1 from 71.5% to 97.8% and K-Means ARI from 0.371 to 0.898. As illustrated in the neighborhood curves (KNN Neighbor Count vs. Macro-F1), scaled configurations persistently dominate unscaled counterparts across every neighborhood size (k).
- **Scale Invariance of Decision Trees:** Tree splits are formed by evaluating features independently through monotonic threshold queries. Because linear feature scaling does not alter the relative rank order of feature values, Decision Tree performance remained strictly identical on both unscaled and scaled representations (0.902 Macro-F1 at depth=3). The hyperparameter curves (Decision Tree Depth vs. Macro-F1) confirm this principle graphically, where the unscaled and scaled validation curves overlap completely.
- **Overfitting Control via Tree Pruning:** Constraining tree depth to max_depth=3 improved validation stability and avoided the performance degradation seen in unconstrained trees (max_depth=None), confirming the necessity of complexity regularization.
- **Supervised vs. Unsupervised Structure:** On scaled data, K-Means achieved an ARI of nearly 0.90 on Wine without accessing ground-truth labels during fitting, proving that the underlying chemical feature space contains clear, naturally separable geometric clusters once variances are normalized.

7. Conclusion:

This benchmark systematically evaluates the impact of feature scaling across supervised and unsupervised algorithms on the Wine and Breast Cancer datasets. The experimental results yield three primary conclusions:

Distance-Based Sensitivity: Distance-dependent models (KNN and K-Means) are heavily compromised by scale disparities. High-magnitude features like *Proline* in the Wine dataset dominate the Euclidean distance metric. Standardizing features to zero mean and unit variance significantly improves performance, boosting KNN Macro-F1 from 0.715 to 0.978 and K-Means ARI from 0.371 to 0.898.

Scale Invariance of Decision Trees: Decision Trees exhibit identical performance under both scaled and unscaled representations (0.902 Macro-F1 on Wine, 0.925 on Breast Cancer). Because decision tree splits rely on monotonic orderings along individual axes rather than geometric distances, feature scaling has no impact on tree topology or decision boundaries.

Generalization and Data Leakage Prevention: By fitting standardizers strictly on the training partitions and applying them downstream, the pipeline ensures unbiased evaluation. On the unseen test partitions, the selected scaled KNN model achieved optimal generalization, yielding an exact 36/36 (100%) accuracy on the Wine dataset and over 0.96 Macro-F1 on the Breast Cancer benchmark.

8. AI-use declaratıon:

Generative AI tools were utilized strictly for structural guidance, code layout formatting, and linguistic refinement of documentation. All underlying analytical logic, experimental executions, hyperparameter decisions, metric interpretations, and live model verifications were conducted, evaluated, and validated independently by the author.

9. References:

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